```
clc;
clear;
close all;
```

1. INPUT PARAMETERS

```
N = 24;                 % FIR Filter Order
M = 2;                  % Decimation Factor
Fs = 8000;              % Sampling Frequency in Hz
N_samples = 128;        % Number of samples

% FIR Prototype Filter
b = fir1(N, 0.4);

% Polyphase Decomposition for M = 2
E0 = b(1:M:end);        % Branch 0 (Even indices)
E1 = b(2:M:end);        % Branch 1 (Odd indices)

% Input Signal: 500 Hz Sinusoid
t = (0:N_samples-1) / Fs;
x = sin(2*pi*500*t);
```

2. CONVENTIONAL VS POLYPHASE DECIMATION

```
% Direct Path: Filtering followed by Downsampling
y_direct_full = filter(b, 1, x);
y_direct = y_direct_full(1:M:end);

% Polyphase Path
x0 = x(1:M:end);
x_delayed = [0, x(1:end-1)];
x1 = x_delayed(1:M:end);

y_poly = filter(E0, 1, x0) + filter(E1, 1, x1);

% Verification Metrics
rec_error = max(abs(y_direct - y_poly));

% Computational Complexity Calculation
N_taps = length(b);
N_in = length(x);
N_out = N_in / M;

MAC_direct = N_in * N_taps;
MAC_poly = N_out * N_taps;
comp_saving = (1 - (MAC_poly / MAC_direct)) * 100;
```

3. COMMAND WINDOW DISPLAY

```
fprintf('============================================================\n');
fprintf(' EXPERIMENT 13: POLYPHASE DECOMPOSITION OF DECIMATION FILTER\n');
fprintf('============================================================\n');

fprintf('Decimation Factor (M) : %d\n', M);
fprintf('Filter Order (N) : %d (%d Taps)\n', N, N_taps);
fprintf('Input Signal : 500 Hz Sinusoid (%d samples)\n', N_samples);
fprintf('Sampling Frequency (Fs) : %d Hz\n\n', Fs);

fprintf('Polyphase Sub-Filter Coefficients:\n');
fprintf(' E_0(z) = [');
fprintf('%.4f ', E0);
fprintf(']\n');

fprintf(' E_1(z) = [');
fprintf('%.4f ', E1);
fprintf(']\n\n');

fprintf('COMPUTATIONAL COMPLEXITY ANALYSIS:\n');
fprintf(' Direct MAC Operations : %d MACs\n', MAC_direct);
fprintf(' Polyphase MAC Operations : %d MACs\n', MAC_poly);
fprintf(' Computational Savings : %.2f%%\n\n', comp_saving);

fprintf('VERIFICATION:\n');
fprintf(' Max Absolute Difference : %.4e\n', rec_error);

if rec_error < 1e-12
    fprintf(' Status : Perfect Reconstruction Verified!\n');
else
    fprintf(' Status : Verification Failed\n');
end

fprintf('============================================================\n');
```

```
============================================================
 EXPERIMENT 13: POLYPHASE DECOMPOSITION OF DECIMATION FILTER
============================================================
Decimation Factor (M) : 2
Filter Order (N) : 24 (25 Taps)
Input Signal : 500 Hz Sinusoid (128 samples)
Sampling Frequency (Fs) : 8000 Hz

Polyphase Sub-Filter Coefficients:
 E_0(z) = [0.0012 0.0000 -0.0072 0.0272 -0.0581 0.0875 0.3989 0.0875 -0.0581
0.0272 -0.0072 0.0000 0.0012 ]
 E_1(z) = [0.0026 -0.0072 0.0112 0.0000 -0.0538 0.2971 0.2971 -0.0538 0.0000
0.0112 -0.0072 0.0026 ]

COMPUTATIONAL COMPLEXITY ANALYSIS:
 Direct MAC Operations : 3200 MACs
 Polyphase MAC Operations : 1600 MACs
```

```
 Computational Savings : 50.00%

VERIFICATION:
 Max Absolute Difference : 3.3307e-16
 Status : Perfect Reconstruction Verified!
============================================================
```

4. SUBFIGURE GENERATION

```
figW = 8.5;
figH = 5.5;

% Subfigure 13A: Prototype Impulse Response
figA = figure('Units', 'centimeters', ...
    'Position', [2, 2, figW, figH]);

stem(0:N, b, 'filled', 'LineWidth', 1.2);
grid on;
box on;
title('Prototype FIR Filter Impulse Response');
xlabel('Sample Index (n)');
ylabel('Coefficient Value');

exportgraphics(figA, 'Exp13_SubfigA_Prototype.png', ...
    'Resolution', 600);

% Subfigure 13B: Sub-filter Coefficients
figB = figure('Units', 'centimeters', ...
    'Position', [11, 2, figW, figH]);

stem(0:length(E0)-1, E0, 'b', 'filled', 'LineWidth', 1.2);
hold on;
stem(0:length(E1)-1, E1, 'r--', 'LineWidth', 1.2);
grid on;
box on;
title('Polyphase Sub-Filter Coefficients (M=2)');
xlabel('Sub-Filter Sample Index');
ylabel('Coefficient Value');
legend('Branch E0[n]', 'Branch E1[n]', 'FontSize', 7);

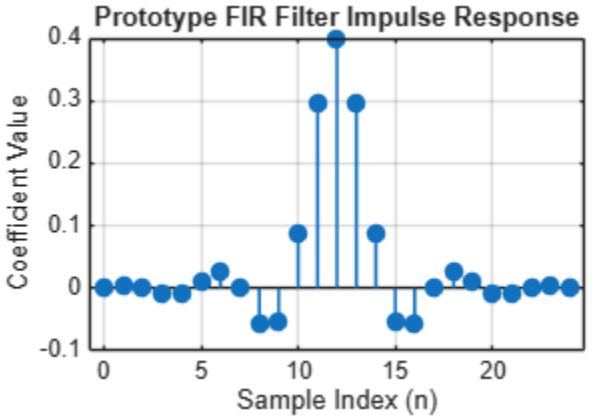
exportgraphics(figB, 'Exp13_SubfigB_Polyphase.png', ...
    'Resolution', 600);

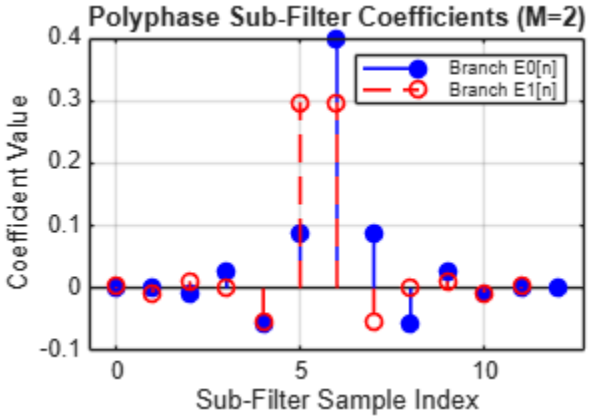
% Subfigure 13C: Time-Domain Output Comparison
figC = figure('Units', 'centimeters', ...
    'Position', [20, 2, figW, figH]);

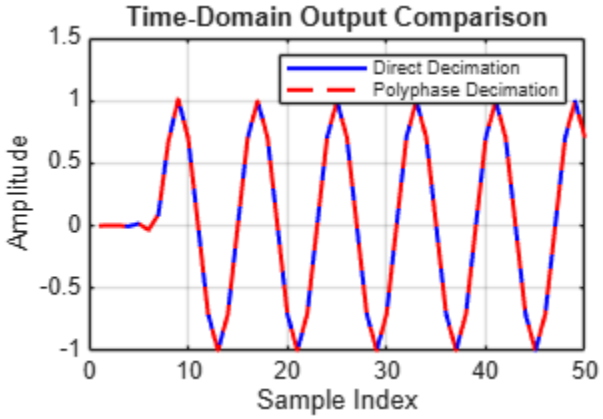
plot(y_direct(1:50), 'b', 'LineWidth', 1.5);
hold on;
plot(y_poly(1:50), 'r--', 'LineWidth', 1.5);
grid on;
box on;
title('Time-Domain Output Comparison');
```

```
xlabel('Sample Index');
ylabel('Amplitude');
legend('Direct Decimation', 'Polyphase Decimation', ...
    'FontSize', 7);

exportgraphics(figC, 'Exp13_SubfigC_OutputComparison.png', ...
    'Resolution', 600);
```







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